

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter 2024****Course: B. Tech.****Branch: Petrochemical Engineering****Subject Name: Unit Operation I****Subject Code: BTPCC302****Semester: III****Date:07/02/2025****Duration: 3.00 Hrs.****Max Marks: 60****Instructions to the Students:**

1. All the questions are compulsory.
2. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Reynolds number is the ratio of..... a. viscous forces to gravity forces. b. inertial forces to viscous forces. c. viscous forces to inertial forces. d. inertial forces to gravity forces.	R	1
2	Manometer is used for measuring----- a. Pressure difference b. Height difference c. Density difference d. Temperature difference	R	1
3	The simple pitot tube measures the _____ pressure. a. Static b. Dynamic c. Total d. none of these	R	1
4	The velocity at which the flow changes from laminar to turbulent is known as a. Mass velocity b. Average velocity c. Critical Velocity d. None of these	R	1
5	Unit Operations are used to conduct a. Physical step of preparing reactant b. Separating and purifying product c. Recycling unconverted reactant d. All of these	R	1
6	Mesh indicates the number of holes per.....	R	1

	a. Square inch	b. Square foot	c. Linear inch	d. Linear foot		
7	The efficiency of the ideal screens is				R	1
	a. Unity	b. Infinity	c. Zero	d. Can't be determined		
8	 is to increase the concentration of solids.				R	1
	a. Thickening	b. Sedimentation	c. Settling	d. Both a & b		
9	The objective of agitation is to				R	1
	a. Provide bulk motion to a liquid	b. Provide Mixing	c. Provide Dispersion	d. All of these		
10	With increase in the temperature, viscosity of a liquid.....				R	1
	a. Increases	b. Decreases	c. Remains constant	d. May increase or decrease; depends on the liquid		
11	The ratio of average fluid velocity to the maximum velocity in case of laminar flow of a Newtonian fluid in a circular pipe is.....				R	1
	a. 0.5	b. 1	c. 2	d. 0.66		
12	Pseudoplastic is an example of -----fluid.				R	1
	a. Newtonian	b. Non-Newtonian	c. Ideal	d. None of the above		
Q. 2	Solve the following.					12
A)	Define Newtonian Fluid, Density, Kinematic viscosity.				R,U	6
B)	Derive an expression for Bernoulli's equation, explain each term in it and also state the assumptions made.				R,U	6
Q.3	Solve the following.					12
A)	Define cavitation. What are some common best practices to prevent cavitation?				R,U,A	6

B)	List various types of valves and mention their specific application in chemical industry.	R,U	6
Q. 4	Solve Any Two of the following.		12
A)	A venturimeter is installed in a pipe line for the measurement of flow rate of water. The pressure drop across the throat and upstream of the meter is ten centimeters of mercury. Calculate the volumetric flow rate of water in m³/s. Data: Diameter of throat = 15 mm Diameter of pipe = 25 mm Coefficient of meter = 0.98 Density of water = 1000 kg/m³ Density of mercury = 13600 kg/m³	U, A	6
B)	Write down difference between variable head meter and variable area meter.	R	6
C)	Define and explain importance of mixing and agitation.	R, U	6
Q.5	Solve Any Two of the following.		12
A)	Write down (each two) industrial application of A) Belt conveyer B) Screw conveyer C) Chain conveyer	U	6
B)	Derive the equation for effectiveness of screens.	R, U	6
C)	Calculate the operating speed of the ball mill from the data given below: Given Data: Diameter of ball mill = 800 mm, diameter of ball = 60mm If (I) operating speed is 55% less than the critical speed. (II) critical speed is 40% more than the operating speed.	U	6
Q. 6	Solve Any Two of the following.		12
A)	What are the good qualities of filter media? What are filter aids? Name few commonly used filter aids.	R, U	6
B)	Describe with neat sketch construction and working of cyclone separator. Write down the industrial applications.	U	6
C)	A six-blade turbine agitator of diameter 60 cm is installed centrally in tank with flat bottom of diameter 180 cm, at a height of 60 cm from the bottom. The tank is filled with a solution of viscosity 10 Cp and of 1.45 g/ml density. The speed of agitation is 90 rpm. The tank is baffled. Calculate the power required. Data: Power number = $N_p = 1.05$ for $N_{Re} > 300$	R, U, A	6
	*** End ***		